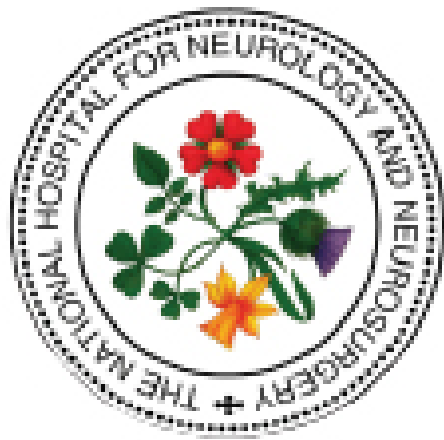




4. DIAGNOSING DEATH by NEUROLOGICAL CRITERIA (DNC) – TEST PROCEDURE

Academy of Medical Royal Colleges (2025) Code of Practice in UK

‘When correctly applied, neurological criteria unequivocally diagnose death, confirming the permanent loss of the capacity for consciousness combined with permanent loss of the capacity to breathe . Certainty that the cessation of brainstem function is permanent is assured by satisfying the criteria below.’

TIMING OF TESTS

Time **before testing** should equal time taken to meet essential preconditions and necessary exclusions - **PRECONDITIONS** and **EXCLUSIONS** must be met before tests are carried out

When decision made that brainstem tests will be completed, Specialist Nurse for Organ Donation (SNOD) should be contacted via 24 hour pager: **03000 20 30 40**

WHO CAN DO THE TESTS

UK Code of Practice requires **two doctors** with expertise in the field to make the diagnosis of brainstem death

- Senior clinicians (registered >5 years) with interest in intensive care medicine / anaesthesia
- One practitioner a consultant, the other consultant / senior registrar
- Testing should be undertaken by both doctors together
- All parts of 1st & 2nd set of tests completed before brainstem death can be confirmed

The two doctors work together to perform full set of clinical tests. Tests are then repeated and must always be performed on two occasions i.e. total of 2 sets of clinical tests (including 2 apnoea tests) performed.

1st set of tests:

DOCTOR A completes set of clinical tests whilst Doctor B observes

2nd set of tests:

DOCTOR B completes set of clinical tests whilst Doctor A observes
- actual tests only performed twice

If 1st set of clinical tests shows no evidence of brainstem function, there need not be a lengthy delay before performing 2nd set.

Appropriate to use time between tests to inform family of result and allow family members to attend 2nd set of clinical tests, if desired.

The patient's arterial blood gases and baseline parameters should be restored to pre-test level, before commencing 2nd set of tests.

TIME OF DEATH

Time of death = time of **2nd set of tests** (or time of ancillary testing if after 2nd set of tests)

TESTS FOR THE ABSENCE OF BRAINSTEM FUNCTION

DIAGNOSTIC TESTS	CRANIAL NERVE <i>S = Sensory M = Motor</i>	AREA OF BRAINSTEM
Pupillary reflex: Pupils fixed midsize or dilated, not reactive to light visible to naked eye or by pupilometer. Direct / consensual response tested on both sides	Optic (II) S Oculomotor (III) M	Midbrain
Corneal reflex: No eyelid movement when each cornea touched in turn – <i>touching sclera not sufficient</i> Care taken to avoid damage to cornea - use sterile gauze	Trigeminal (V) S Facial (VII) M	Midbrain
Vestibulo-ocular reflex – No eye movement during or following slow injection of ≥50ml ice-cold water over 1min into each external auditory meatus in turn with head flexed at 30° <i>Prior to test, clear access to tympanic membrane established by direct inspection</i>	Oculomotor (III) M Trochlear (IV) M Abducens (VI) M Acoustic (VIII) S	Pons
No motor response within cranial nerve or somatic distribution to stimuli No limb response to supraorbital nerve pressure – performed on both sides Absent grimacing to pain No head movement <i>Somatic reflex limb and trunk movements (spinal reflexes) differentiated from brain mediated function – if doubt, ancillary investigation required</i>	Trigeminal (V) S Facial (VII) M Accessory (XI) M	Midbrain Medulla
No gag / cough reflex in response to suction catheter passed down trachea, no slowing of heart rate	Glossopharyngeal (IX) S Vagus (X) S/M	Medulla

APNOEA TEST *

Patients can appear to be spontaneously breathing while connected to mechanical ventilator due to ventilator auto-cycling/triggering
Apnoea test must not be performed while connected to ventilator

1. Pre-oxygenate with FiO₂ 1.0 /10min
2. Check ABG to confirm measured P_aCO₂ & S_aO₂ correlate with monitored values
3. Maintain SaO₂ >95%, reduce SIMV rate to allow slow rise in E_TCO₂
4. Check ABG to confirm starting PaCO₂ ≥5.3kPa
5. DISCONNECT from ventilator
6. Attach to MAPLESON C WATERS CIRCUIT with 1.0 FiO₂
7. **AUGMENTED CARBON DIOXIDE APNOEA TEST**
8. Check ABG to confirm that:
 - PaCO₂ ≥8.0kPa
 - PaCO₂ risen by 2.7kPa from starting PaCO₂
 - Arterial pH <7.3 ([H⁺] >50 nmol/L)
10. Observe for minimum 5min to confirm no spontaneous respiration
11. Reconnect to ventilator - *In case of absolute inability to complete apnoea test, for whatever reason, death using neurological criteria can only be made by confirming both absence of brainstem reflexes + addition of ancillary investigation*

Start: PaCO₂ ≥5.3kPa
If hard to maintain adequate PaO₂ use CPAP with Waters circuit

Time. Observed for minimum 5min

Target. PaCO₂ must be ≥8.0kPa

End: PaCO₂ 2.7kPa >**Start** PaCO₂

* **CERVICAL SPINAL CORD INJURY** - If evidence of injury do not proceed with apnoea test since injury renders its interpretation invalid

a) Upper cervical spinal cord injury alone may cause apnoea

b) Cervical cord injury must be excluded where brain injury caused by trauma

c) Ancillary investigations required

As PaCO₂ rises there will be increase in precordial (cardiac) and abdominal (aortic) pulsations and should not be confused with respiratory effort - watch reservoir bag for any movement to identify respiratory effort